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End Semester Examination 2025

Name of the Course: B.Tech (ECE)
Name of the Paper: Satellite Communications
Time: 3 Hours

Semester: VIII Sem
Paper Code: **TEC 851**
Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	What is the difference between geosynchronous and a geostationary satellite? Discuss about frequency allocations for satellite services.	CO 1
(b)	Explain Kepler's three laws and their significance in satellite communications. Discuss the effects of orbital perturbations on satellite performance.	
(c)	Determine the limits of visibility for an earth station situated at mean sea level, at latitude 48.42 degrees north and longitude 89.26 degrees West. Assume a minimum angle of elevation of 5 degrees.	
Q2	(20 marks)	
(a)	Explain the importance of telemetry, tracking, and command (TTC & M) in satellite operations.	CO 2
(b)	List the sub-systems of satellite communication. Describe the function and working of the Attitude and Orbit Control System (AOCS).	
(c)	A Satellite at a distance of 40,000 km from a point on earth's surface, radiates a power of 200 mW from an antenna with a gain of 17 dB in the direction of the observer. Find the flux density at the receiving point and the power received by an antenna with an effective area of 10 m ² .	
Q3	(20 marks)	
(a)	Derive the general satellite link design equation and explain each term.	CO 3
(b)	Discuss how rain attenuation and depolarization affect satellite communication links.	
(c)	A satellite stationed at a distance 38000 km from the surface of the earth radiates a power of 5watts, in the direction of an earth station. Assuming the satellite antenna gain to be 30 dB. If the operating frequency is 10 GHz and gain of receiving antenna is 70 dB calculate the power received.	
Q4	(20 marks)	
(a)	Differentiate between FDMA and TDMA with respect to bandwidth and power efficiency.	CO 4
(b)	Explain the principle behind spectrum spreading and despreading and how this is used to minimize interference in a CDMA system. Also determine the throughput efficiency of the system	
(c)	If a satellite has a transponder having 50 MHz BW and 1023 barker code is used for CDMA by various earth stations, the chip rate is 16 MCPS. Determine the number of accesses for an SNR of 16 dB. Also find the total occupancy of transponder if BPSK is used.	

Q5	(20 marks)	CO 5, 6
(a)	What is meant by DTH? What are the design issues to be considered for launching DTH systems?	
(b)	Write a short note on (a) GPS Navigation Message (b) GPS Signal Levels.	
(c)	Discuss the basic principles of GPS operation based on the concept of trilateration. Explain two major parameters that cause errors in positioning.	